Recap.

① $p_\theta \approx p_{data}$: unknown.

$$\theta^* \in \arg\min_\theta D(p_{data} \| p_\theta).$$

$$D_{KL}(p_{data} \| p_\theta) = \int p_{data}(w) \cdot \log \frac{p_{data}(w)}{p_\theta(w)} \, dw$$

$$= \mathbb{E}_{w \sim p_{data}} [\log p_{data}(w) - \log p_\theta(w)]$$

- $D_{KL} \geq 0$.
- $D_{KL} = 0$ iff $p_{data} = p_\theta$ a.s.

$$\min_\theta D_{KL}(p_{data} \| p_\theta) \Leftrightarrow \max_\theta \mathbb{E}_{w \sim p_{data}} [\log p_\theta(w)]$$

$$\Rightarrow \max_\theta \frac{1}{N} \sum_{i=1}^{N} \log p_\theta(w^{(i)}), \quad w^{(i)} \sim p_{data}$$

$$p_\theta(w) = \prod_{t=1}^{n} p_\theta(w_t \mid w_{1:t-1})$$

② N-gram : $p_\theta(w_t \mid w_{1:t-1}) \approx p_\theta(w_t \mid w_{t-N+1:t-1})$

$$MLE = \frac{c(w_{1:N})}{c(w_{1:N-1})} \text{ : history}$$

③ PPL (perplexity) $D_{test} = \{s_{1:m}\}$

$$= 2^{-l}, \quad l = \frac{1}{\sum_{i=1}^{m} |s_i|} \sum_{i=1}^{m} \log_2 p_\theta(s_i)$$

$$= \left[\prod_{i=1}^{m} p_\theta(s_i)\right]^{-\frac{1}{\sum_{i=1}^{m} |s_i|}}$$

④ kN-smoothing. / ∞-gram.

① . classification: $\begin{cases} NB \\ LR \end{cases}$

NB: $c_{MAP} := \arg\max_{c \in C} p(c|d) := \frac{p(d|c) \cdot p(c)}{p(d)}$

$:= \arg\max_{c \in C} p(d|c) \cdot p(c)$ $\quad \theta = [\hat{p}(c), \hat{p}(v_i|c)]$.

LR: $z = w^T x + b$: directly models $p_\theta(y|x)$.

$\begin{cases} z \uparrow \to \text{positive} \\ z \downarrow \to \text{negative} \end{cases}$ $\quad x \to \theta \to \sigma \to p_\theta$

z : $\sigma(z) = \frac{1}{1+e^{-z}}$, $1-\sigma(z) = \frac{1+e^{-z}-1}{1+e^{-z}} = \frac{1}{1+e^{z}} = \sigma(-z)$

$1-\sigma(z) = \sigma(-z)$.

$\theta = [w_1, \cdots, w_p, b]$.

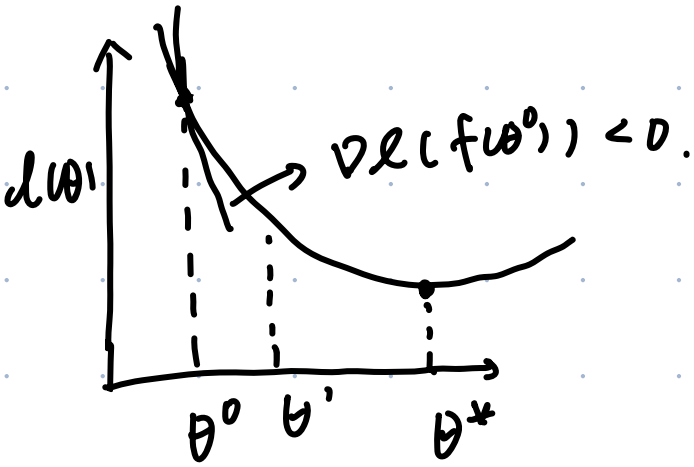
softmax: $\text{softmax}(z_c) = \frac{\exp(z_c)}{\sum_{j=1}^{k} \exp(z_j)}$

$z_c = w_c^T \cdot x + b_c$, $\quad \theta = \begin{bmatrix} w_1, b_1 \\ w_2, b_2 \\ \vdots \\ w_{k-1}, b_{k-1} \end{bmatrix}$

$p_\theta(y|x) = \hat{y}^y (1-\hat{y})^{1-y}$ $\quad \begin{cases} y=1, \ p_\theta(y|x) = \hat{y} \\ y=0, \ p_\theta(y|x) = 1-\hat{y} \end{cases}$

It is natural to maximize $p_\theta(y|x)$

$H(p,q) := -\sum_{x \in X} p(x) \log q(x)$



$$\theta' = \theta^0 - \eta \cdot \nabla l(f(\theta^0))$$

η: step size / learning rate.

- generative model
- discriminative model

②. Counting based:

1. TF-IDF:

$$TF_{t,d} = \log_{10}(\text{count}(t,d)+1)$$

$$DF_t = \#\{\text{docs containing } t\}.$$

$$IDF_t = \log_{10}\left(\frac{N}{DF_t}\right)$$

$$w_{t,d} = TF_{t,d} \cdot IDF_t.$$

2. PPMI: $p_{i,j} = \dfrac{f_{ij}}{\sum_{i=1}^{|V|}\sum_{j=1}^{C} f_{ij}}$, $p_{i*} = \dfrac{\sum_{j=1}^{C} f_{i,j}}{\sum_{i=1}^{|V|}\sum_{j=1}^{C} f_{ij}}$

$$\boxed{f_{ij}: \#\text{ times } w_i \text{ occurs with context } c_j.}$$

$$p_{*j} = \frac{\sum_{i=1}^{|V|} f_{ij}}{\sum_{i=1}^{|V|}\sum_{j=1}^{C} f_{ij}}$$

$$PMI(w_1,w_2) = \log \frac{p(w_1,w_2)}{p(w_1)\cdot p(w_2)}$$

$$PPMI(w_1, w_2) = \max\left(\log_2 \frac{p_{i,j}}{p_{i*} \cdot p_{j*}}, 0\right).$$

③. learning-based method: $U = [u_1, u_2, \dots, u_{|V|}]$

$V = [v_1, v_2, \dots, v_{|V|}]$

wordzvec:

- Positive pairs:

$$p(D=1 \mid w, c) = \sigma(u_w^T \cdot v_c)$$

c is a true context word of w

- Negative pairs:

$$p(D=0 \mid w, c) = 1 - \sigma(u_w^T \cdot v_c)$$
$$= \sigma(-u_w^T \cdot v_c)$$

c is NOT a true context word of w.

How to get non-context c in practice and how many of them?

c_{pos} : $c_{neg_1}, c_{neg_2}, \dots, c_{neg_k}$

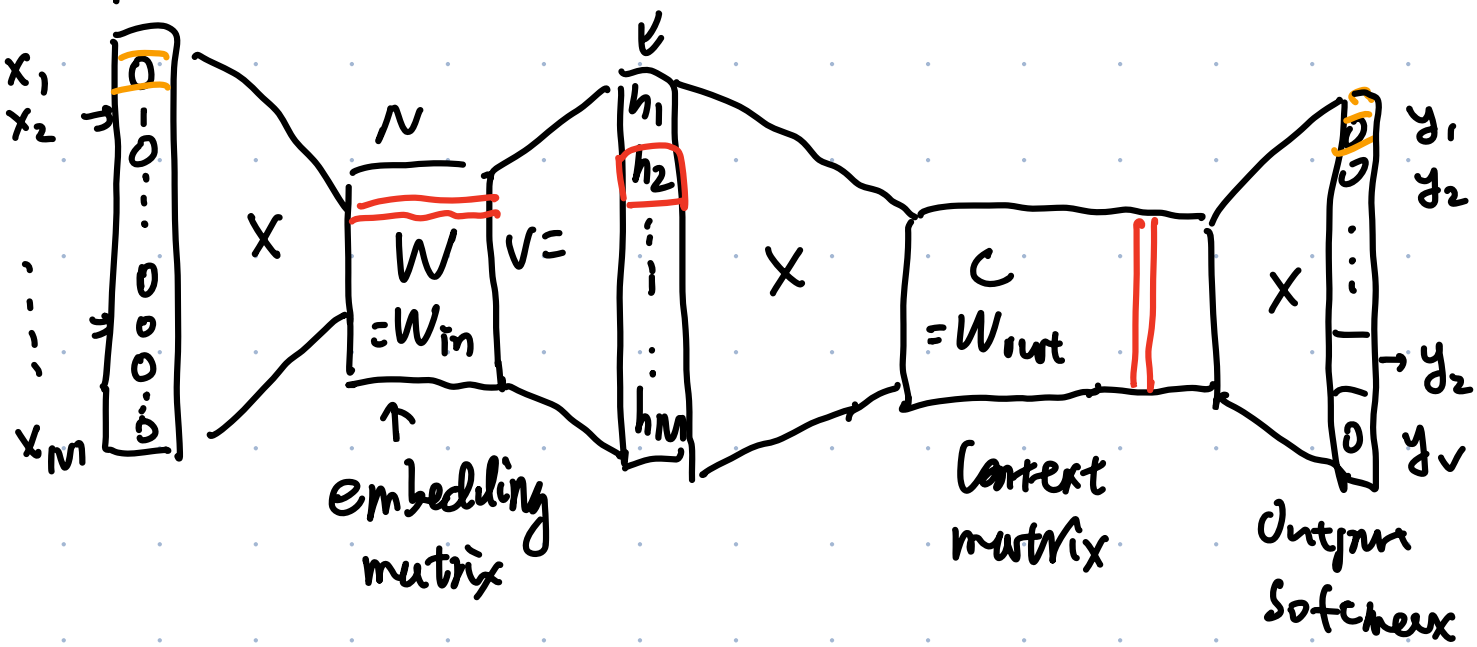
$$MLE(w, c_{pos}, c_{neg_{1:k}}) = \log\left\{p(D=1 \mid w, c_{pos}) \cdot \prod_{i=1}^{k} p(D=0 \mid w, c_{neg_i})\right\}.$$

$\Leftrightarrow$ minimize CE:

$$\min L := -\left[\log \sigma(w^T \cdot c_{pos}) + \sum_{i=1}^{k} \log \sigma(-w^T \cdot c_{neg_i})\right]$$

NN:

$\theta = [W_{in}, W_{out}]$



$\Leftrightarrow CE(y_{pred}, y_{true})$

$$y_{pred} = \frac{\exp(W_{out}(c_{posi}) \cdot h)}{\underbrace{\sum_{i=1}^{|V|} \exp(W_{out(i)} \cdot h)}_{\text{too expensive.}}}$$

$$\arg\max_{\theta} \prod_{w \in Text} \left[\prod_{c \in C(w)} p(c \mid w; \theta) \right]$$

$$\Leftrightarrow \arg\max_{\theta} \prod_{(w,c) \in D} p(c \mid w; \theta)$$

$\Rightarrow$ adding negative sampling:

$$J(\theta) := -\frac{1}{T}\sum_{t=1}^{T}\sum_{-k\le j\le k,\, j\ne 0} \log p(c_{t+j} \mid w_t; \theta)$$

$$\theta^{t+1} = \theta^{t} - \eta_t \nabla J(\theta)$$

$$\eta_t = \eta_0 \cdot \left(1 - \frac{wc}{T' \cdot |V| + 1}\right)$$

wc : running # words processed.

1. window : 5-20.
2. negative samples : $\begin{cases} k=2\text{-}5 : \text{large data sets} \\ k=5\text{-}20 : \text{small datasets} \end{cases}$
3. draw negative samples:

$P_{|V|}(w)$:

- $U(w)$: # in corpus
- Z : total tokens in text
- $U(w)/Z$:
- $P_{|V|}(w) = \left(\frac{U(w)}{Z}\right)^{\alpha}$.
- $\alpha = 0.75$